

Week 5

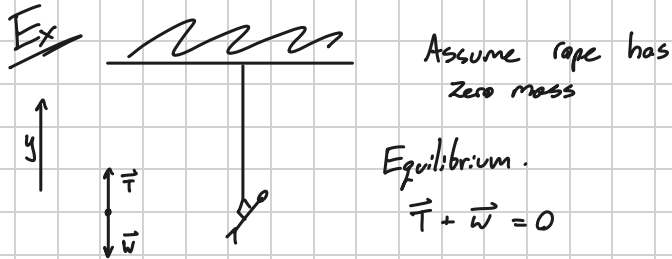
Newton's first Law

→ Particle in equilibrium

Condition for equilibrium $\sum \vec{F} = 0$

in 2D $x: \sum F_x = 0$

$y: \sum F_y = 0$



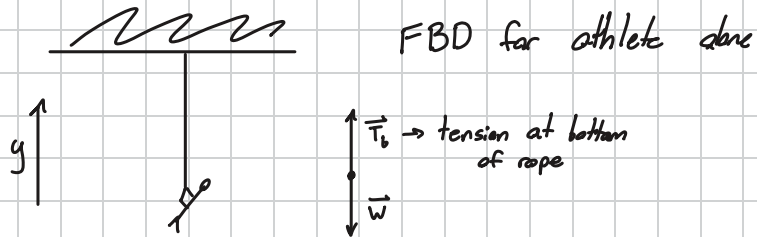
Equilibrium.

$$\vec{T} + \vec{W} = 0$$

$y: W = -mg$

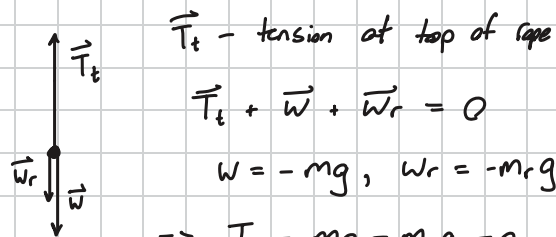
$$T - mg = 0 \Rightarrow T = mg$$

~~Ex~~ rope has mass m_r



$$T_b - mg = 0 \Rightarrow T_b = mg$$

FBD for athlete + rope system

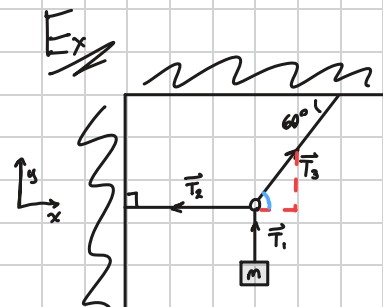


$$\vec{T}_t + \vec{w} + \vec{w}_r = 0$$

$$w = -mg, w_r = -m_r g$$

$$\Rightarrow T_t - mg - m_r g = 0 \Rightarrow$$

$$T_t = (m + m_r)g$$



the string and ring are massless. find tension.

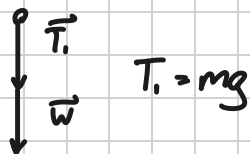
Forces on ring: $\vec{T}_1, \vec{T}_2, \vec{T}_3$

Ring is in equilibrium: $\vec{T}_1 + \vec{T}_2 + \vec{T}_3 = 0$

$$x: -T_2 + T_3 \cos \theta = 0 \quad (1)$$

$$y: -T_1 + T_3 \sin \theta = 0 \quad (2)$$

FBD for mass m:



Sub into (2): $-mg + T_3 \sin \theta = 0$

$$\Rightarrow T_3 = \frac{mg}{\sin \theta}$$

Now Sub into ①

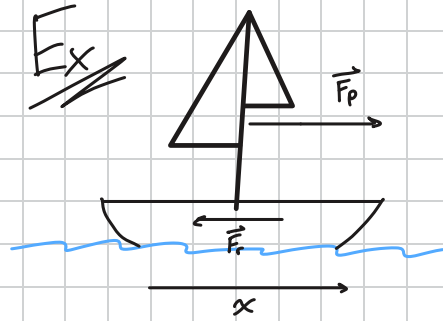
$$-T_2 + \frac{mg}{\sin\theta} \cdot \cos\theta = 0$$

$$\Rightarrow T_2 = \frac{mg \cos\theta}{\sin\theta} = mg \cot\theta$$

Newton's Second Law

$$\sum \vec{F} = m\vec{a}$$

$$\begin{aligned} \text{2D problem} \Rightarrow \quad X: \sum F_x &= ma_x \\ Y: \sum F_y &= ma_y \end{aligned}$$



$$M = 200 \text{ kg}$$

$$F_r = 100 \text{ N}$$

$$a_x = 1.5 \text{ m/s}^2$$

What is F_p ?

$$X: F_p - F_r = ma_x \Rightarrow F_p = F_r + ma_x \Rightarrow F_p = 400 \text{ N}$$

